

ORANGE CITY, IA, USA

WMO: 725489

Lat: 42.990N

Lon: 96.063W

Elev: 1414

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-3.6	-15.3	2.5	-6.8	-10.6	3.3	-1.4	32.5	20.2	28.0	22.3	6.9	300	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	91.2	75.0	89.7	74.0	86.2	72.7	80.0	86.4	77.8	84.7	75.5	83.3	12.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	156.6	84.4	75.2	139.6	82.7	72.8	128.4	80.8	44.8	86.4	42.2	85.0	40.2	83.2	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.7	20.1	DB	-15.1	96.0	5.5	3.2	-19.1	98.3	-22.3	100.2	-25.5	101.9	-29.5	104.2	(4)
(5)			WB	-14.9	82.6	6.1	2.9	-19.3	84.7	-22.8	86.4	-26.3	88.0	-30.7	90.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	19.9	23.0	36.2	49.1	60.6	70.8	74.5	71.9	64.7	51.0	36.7	23.0	(6)	
	DBStd	21.84	12.64	12.27	12.52	10.19	9.02	6.79	5.96	5.42	8.46	9.69	10.77	11.78	(7)	
	HDD50	3657	934	757	454	137	14	0	0	0	4	105	415	837	(8)	
	HDD65	6894	1399	1177	893	485	195	25	5	9	109	445	850	1302	(9)	
	CDD50	3138	0	0	27	109	345	624	759	679	445	136	14	0	(10)	
	CDD65	899	0	0	1	6	60	199	299	223	100	11	0	0	(11)	
	CDH74	8280	0	0	11	114	629	1878	2872	1765	886	122	3	0	(12)	
	CDH80	2820	0	0	1	29	209	668	1072	535	277	29	0	0	(13)	
(14) Wind	WSAvg	8.8	9.4	9.5	10.0	10.7	10.1	8.7	6.7	6.3	7.5	8.8	9.4	9.2	(14)	
(15) Precipitation	PrecAvg	28.9	0.7	0.7	1.7	3.0	3.5	4.4	3.7	3.5	3.3	2.3	1.2	0.9	(15)	
	PrecMax	38.3	1.3	1.2	4.4	5.0	7.1	9.1	9.2	8.3	8.9	5.6	3.7	2.3	(16)	
	PrecMin	22.2	0.1	0.0	0.0	0.7	1.1	1.1	0.2	0.9	0.8	0.6	0.0	0.0	(17)	
	PrecStd	4.3	0.4	0.3	1.1	1.1	1.5	1.8	2.2	1.9	1.8	1.6	1.0	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	54.6	74.9	83.9	91.4	93.5	94.9	92.6	91.2	84.0	70.1	53.7	(19)	
		MCWB	43.6	46.6	56.7	60.4	67.9	73.6	77.9	74.9	70.5	63.7	56.1	47.5	(20)	
	2%	DB	44.6	46.5	67.9	76.8	85.2	90.4	91.2	89.6	86.4	77.0	63.1	45.8	(21)	
		MCWB	38.9	41.4	55.1	58.7	66.1	72.4	77.5	75.7	70.1	60.5	52.6	40.1	(22)	
	5%	DB	38.7	42.7	61.1	71.8	81.2	87.5	89.8	85.7	82.3	72.1	56.6	40.9	(23)	
		MCWB	35.5	38.6	52.8	56.1	63.2	71.2	76.8	75.5	67.9	58.4	47.7	37.1	(24)	
	10%	DB	36.1	38.5	54.5	65.9	76.6	83.8	86.1	82.1	79.3	66.3	52.0	37.0	(25)	
		MCWB	33.9	35.6	47.5	53.5	61.4	69.4	75.1	72.4	66.2	56.6	45.4	34.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	47.1	63.0	64.4	74.5	78.7	84.4	82.2	76.5	70.5	59.3	48.9	(27)	
		MCDB	51.0	52.4	67.6	76.2	83.3	88.3	90.7	86.7	84.3	77.3	67.6	52.2	(28)	
	2%	WB	38.9	42.2	57.5	61.0	69.1	76.2	81.2	79.3	74.2	64.3	53.3	40.9	(29)	
		MCDB	42.8	46.7	63.1	72.7	80.1	85.4	87.2	85.4	81.7	71.8	61.0	45.6	(30)	
	5%	WB	36.1	38.7	52.8	57.7	66.3	74.1	78.5	77.0	71.7	60.7	49.2	37.6	(31)	
		MCDB	38.6	42.4	60.6	68.2	76.7	83.5	85.4	82.9	79.7	68.9	55.5	40.4	(32)	
	10%	WB	33.7	35.8	47.5	54.9	63.7	71.6	76.4	74.5	69.2	56.8	45.2	35.2	(33)	
		MCDB	35.7	38.5	55.1	64.4	72.4	80.5	84.0	81.0	76.4	64.7	50.2	37.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.2	18.1	20.8	20.0	19.1	19.0	18.4	21.0	19.9	17.8	14.6	(35)	
		MCDBR	20.6	20.1	28.9	29.5	26.1	22.3	20.6	21.4	24.6	26.7	25.5	19.5	(36)	
	5% WB	MCWBR	16.4	15.3	17.5	15.1	12.0	10.3	11.1	12.0	12.0	14.1	15.9	15.1	(37)	
		MCWBR	18.9	18.7	26.0	25.4	21.5	18.9	18.2	18.6	20.2	22.3	23.7	18.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.273	0.284	0.318	0.369	0.393	0.400	0.410	0.408	0.362	0.328	0.291	0.277	(40)	
		taud	2.431	2.421	2.412	2.292	2.297	2.335	2.300	2.293	2.395	2.468	2.491	2.443	(41)	
	Ebn at Noon		278	293	294	285	279	277	272	269	275	272	268	265	(42)	
		Edh at Noon	23	28	33	40	41	39	40	39	33	27	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		607	948	1219	1537	1786	1995	2098	1748	1406	962	664	491	(44)	
	RadStd		56	75	97	131	130	142	95	126	95	118	42	41	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page